Lab 8

Implementing AI Models in Prolog

Why Use Prolog for AI?

Prolog is a **logic-based** programming language that excels at:

- **Symbolic reasoning**
- **Knowledge representation**
- **Rule-based expert systems**
- **Automated decision-making**

Unlike Python (which is widely used for ML models), Prolog is ideal for problems that involve **logical inference, search, and constraint satisfaction**.

AI Techniques in Prolog

1. **Expert Systems**
   - Knowledge-based systems that **simulate human experts**.
   - Example: Medical diagnosis systems.
2. **State-Space Search**
   - Used in pathfinding, problem-solving (e.g., **Water Jug Problem**).
3. **Constraint Satisfaction Problems (CSPs)**
   - Solves problems like **Sudoku, N-Queens, and Scheduling**.
4. **Natural Language Processing (NLP)**
   - Prolog can be used to build **parsers and chatbots**.

Example 1: Expert System in Prolog (Simple Medical Diagnosis)

```
% Facts
symptom(john, fever).
symptom(john, cough).
symptom(john, sore_throat).

% Rules
disease(X, flu) :- symptom(X, fever), symptom(X, cough), symptom(X,
sore_throat).
disease(X, cold) :- symptom(X, cough), symptom(X, runny_nose).

% Query
?- disease(john, What).
% Output: What = flu.
```

Example 2: AI Search Algorithm (Depth-First Search in Prolog)

```
% Graph representation
edge(a, b).
edge(b, c).
edge(c, d).
edge(b, e).

% Depth-First Search
dfs(Start, Goal, Path) :- dfs_helper(Start, Goal, [Start], Path).

dfs_helper(Goal, Goal, Path, Path).
dfs_helper(Node, Goal, Visited, Path) :-
    edge(Node, Next),
    \+ member(Next, Visited),
    dfs_helper(Next, Goal, [Next | Visited], Path).

% Query
?- dfs(a, d, Path).
% Output: Path = [d, c, b, a].
```

Propositional Logic in AI

What is Propositional Logic?

Propositional Logic (also called **Boolean Logic**) is a **formal system** in logic that uses **propositions (statements)** that can be **true (T) or false (F)**.

It is **widely used in AI** for:

- **Knowledge representation** (facts and rules).
- **Automated reasoning** (inference engines).
- **Decision-making systems** (expert systems).

Basic Components of Propositional Logic

1. **Propositions (Statements):**
   - **Atomic propositions:** Simple statements. Example:
     - *"It is raining."*
     - *"The sky is blue."*
   - **Compound propositions:** Formed by combining atomic propositions using **logical connectives**.
2. **Logical Connectives:**
   - **AND (∧)** → True if both propositions are true.
   - **OR (∨)** → True if at least one proposition is true.
   - **NOT (¬)** → Negates the truth value.
   - **IMPLICATION (→)** → "If P, then Q."
   - **BICONDITIONAL (↔)** → "P if and only if Q."

Examples of Propositional Logic Statements

Statement	Propositional Logic Expression
"It is raining and cold."	$R \wedge C$
"It is raining or cold."	$R \vee C$
"If it is raining, then the ground is wet."	$R \rightarrow W$
"The ground is wet if and only if it is raining."	$W \leftrightarrow R$
"It is not raining."	$\neg R$

Truth Tables in Propositional Logic

A **truth table** shows all possible values of a logical expression.

Example: AND (P ∧ Q)

P	Q	P ∧ Q
T	*T*	*T*
T	*F*	*F*
F	*T*	*F*
F	*F*	*F*

Example: OR (P ∨ Q)

P	Q	P ∨ Q
T	*T*	*T*
T	*F*	*T*
F	*T*	*T*
F	*F*	*F*

Propositional Logic in Prolog

Prolog can be used to **implement propositional logic** using **facts and rules**.

Example: Defining Propositional Logic in Prolog

```
% Facts
raining.
cold.

% Rules
wet_ground :- raining.
wear_jacket :- cold.

% Query
?- wet_ground.   % Output: true
?- wear_jacket.  % Output: true
```

Logic Programming in AI

What is Logic Programming?

- **Logic Programming** is a programming paradigm based on **formal logic**.
- A program consists of **facts**, **rules**, and **queries**.
- Prolog (Programming in Logic) is the most used **logic programming language**.

Key Concepts of Logic Programming

1. **Facts:**
   - Basic statements that are always true.
   - Example:

```
cat(whiskers).
dog(buddy).
```

2. **Rules:**
   - Derived facts based on conditions.
   - Example:

```
mammal(X) :- cat(X).
mammal(X) :- dog(X).
```

3. **Queries:**
   - Asking the system for information.
   - Example:

```
?- mammal(whiskers).  % Output: true
?- mammal(buddy).     % Output: true
```

Example: Family Tree in Prolog

```
% Facts
parent(john, mary).
parent(mary, alice).

% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Queries
?- grandparent(john, alice).  % Output: true
```

Hybrid AI Systems: Integrating Prolog with Python

Why Integrate Prolog with Python?

- Prolog is great for **logic and reasoning**, while Python is excellent for **data handling and AI models**.
- Combining them allows us to build **hybrid AI systems** that **leverage logic and machine learning**.

Method 1: Using PySwip (Python Interface for Prolog)

1. Install PySwip

```
pip install pyswip
```

2. Python Code to Run Prolog Queries

```
from pyswip import Prolog

prolog = Prolog()
prolog.assertz("parent(john, mary)")
prolog.assertz("parent(mary, alice)")

result = list(prolog.query("parent(john, X)"))
print(result)  # Output: [{'X': 'mary'}]
```

Method 2: Calling Python from Prolog

- Prolog can **call Python functions** using external interfaces.

Example: Calling a Python Function from Prolog

```
:- use_module(library(python)).

run_python_function :-
    python_call(print("Hello from Python!")).
```

Lab 8 - Tasks

Task 1; Create any of the following, can use Free API & develop its "Flask" based back-end + font-end (make it dynamic aswell):

1. Weather - App
2. Cooking / Recipe - App
3. Spotify - App
4. News - App
5. Vehicle Info - App
6. Useless Info - App
7. Random Joke - App
8. Stock Price - App
9. NASA - App
10. Your Own Choice - App